

QUALITY ASSURANCE & BIOSAFETY COMPLIANCE DIVISION

Microbiology Culture Safety Audit

Facility: Meridian BioLabs Central Containment Hub (Building C)

98.4%

OVERALL COMPLIANCE

ZERO

CRITICAL DEFICIENCIES

2 ITEMS

ACTION ITEMS PENDING

142

LOGGED CULTURES

Audit Purpose & Scope: This audit evaluates microbiological safety, containment protocols, pathogen storage logs, and equipment validation for Meridian BioLabs' BSL-2 and BSL-3 suites in accordance with CDC/NIH *Biosafety in Microbiological and Biomedical Laboratories* (BMBL 6th Edition) and ISO 15189 standards.

1. Biosafety Containment & Facility Operations Checklist

ID	Inspection Criteria & Standard Ref.	Suite	Status	Auditor Observations & Notes
C-01	Negative Pressure Differential (≥ 30 Pa relative to hallway)	BSL-3	✓ COMPLIANT	Continuous pressure monitoring active. Digital sensors calibrated June 2026.
C-02	Class II Type A2 Biosafety Cabinet Laminar Velocity (100 fpm)	BSL-2	✓ COMPLIANT	All 6 cabinets certified by Vertex Air Systems. Next service due Jan 2027.
C-03	Autoclave Efficacy Validation (<i>Geobacillus stearothermophilus</i>)	Shared	✓ COMPLIANT	Weekly spore strip tests verified clear. Sterilization logs complete.
C-04	Emergency Eyewash & Safety Shower Weekly Flush Logs	BSL-2/3	! ACTION REQUIRED	Eyewash unit #3 in Room B3-102 missing 2 signature entries for early July.
C-05	High-Efficiency Particulate Air (HEPA) Exhaust Filter Integrity	BSL-3	✓ COMPLIANT	DOP smoke test passed with 99.997% efficiency on primary filter bank.
C-06	Access Control & Bio-containment Key-card Security Logs	BSL-3	✓ COMPLIANT	Dual-factor biometric entry log reconciled. Zero unauthorized attempts.
C-07	Chemical & Biological Hazard Emergency Spill Kit Readiness	BSL-2/3	! ACTION REQUIRED	BSL-2 main lab spill kit missing fresh 10% bleach concentrate solution.
C-08	Waste Segregation & Sharps Containment Disposal Protocol	BSL-2	✓ COMPLIANT	Biohazard bins properly color-coded and sealed prior to incineration.

2. Microorganism Culture Repository Inventory Log (Sample Audit)

Strain ID	Microorganism Species	Risk	Storage Location	Vials	State	Authorized Custodian	
MB-ECOLI-01	<i>Escherichia coli</i> K-12	RG1	Cryo-Freezer (−80°C)	A1	24	Frozen	Dr. Marcus Vance
MB-STAPH-04	<i>Staphylococcus aureus</i> ATCC 25923	RG2	Cryo-Freezer (−80°C)	B2	12	Frozen	Dr. Sarah Jenkins
MB-PSEUD-09	<i>Pseudomonas aeruginosa</i> PAO1	RG2	Liquid N2 Tank 01	8	Cryo	Dr. Marcus Vance	
MB-CANDI-02	<i>Candida albicans</i> SC5314	RG2	Incubator 3 (37°C)	4	Slant	Tech. David Miller	
MB-ASPER-01	<i>Aspergillus fumigatus</i> Af293	RG2	Cryo-Freezer (−80°C)	B1	6	Spore	Dr. Sarah Jenkins
MB-MYCO-03	<i>Mycobacterium smegmatis</i> MC2 155	RG1	Cryo-Freezer (−80°C)	A2	15	Frozen	Tech. Elena Rostova
MB-SALM-11	<i>Salmonella enterica</i> ser. Typhimurium	RG2	Liquid N2 Tank 02	10	Cryo	Dr. Marcus Vance	
MB-BACIL-02	<i>Bacillus subtilis</i> 168	RG1	Storage Cabinet C	30	Dry	Tech. David Miller	
MB-BRUCL-01	<i>Brucella abortus</i> RB51 (Vaccine)	RG3	BSL-3 Vault-Freezer 01	4	Cryo	Dr. Eleanor Vance	
MB-VIBRI-05	<i>Vibrio cholerae</i> O1 El Tor	RG2	Cryo-Freezer (−80°C)	B3	6	Frozen	Dr. Sarah Jenkins